



**Faculty of Computer Applications &
Information Technology**

iMScIT Programme Semester VI

221601602
Machine Learning

Module - 1

**RELATIONSHIP
BETWEEN
ARTIFICIAL
INTELLIGENCE,
MACHINE
LEARNING
DEEP
LEARNING
AND
DATA SCIENCE**

Artificial Intelligence (AI)

Programs with the ability to learn and reason like humans

Machine Learning (ML)

Algorithms with the ability to learn without being explicitly programmed

Deep Learning (DL)

Subset of machine learning in which artificial neural networks adapt and learn from vast amounts of data

Data Science

Data Science is a field of study that combines statistics & maths, programming skills - Python, R etc. and domain expertise to extract meaningful insights from data.

Maths & Statistics

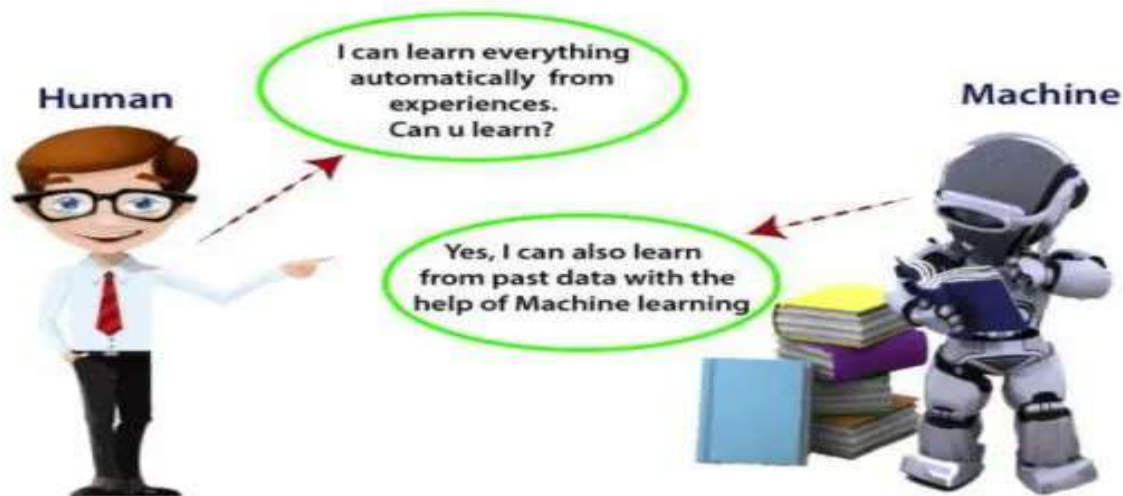
Domain Knowledge

Programming Skills

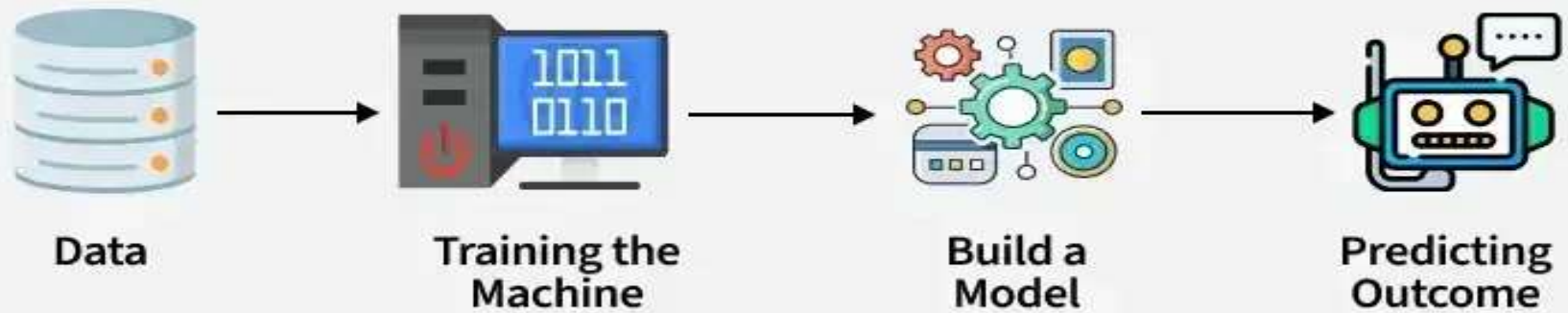
Data Science integrates all the above terms - AI, ML & DL to extract insights from data (exploratory data analysis) and make predictions from large datasets (predictive analytics).

What is Machine Learning?

- Machine learning is a branch of artificial intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy.



Introduction To Machine Learning



Why is Machine Learning Important?

- Machine Learning can **reduce costs**, **mitigate risks**, and improve quality of life by recommending products/services, **detecting cybersecurity breaches**, and enabling **self-driving cars**. It is becoming more common and will soon **integrate** into many facets of life.
- Machine Learning is a popular subfield of Artificial Intelligence used in various fields, including **healthcare**, **finance**, **infrastructure**, **marketing**, **self-driving cars**, recommendation systems, **chatbots**, **social sites**, **gaming**, **cyber security**, and others.
- Machine Learning is critical because it allows businesses to interpret customer behavior trends and understand business operation patterns in a broader context. Furthermore, today's top companies, such as **Facebook**, **Google**, and **Uber**, are prioritizing Machine Learning in their operations.

Why do we need Machine Learning?

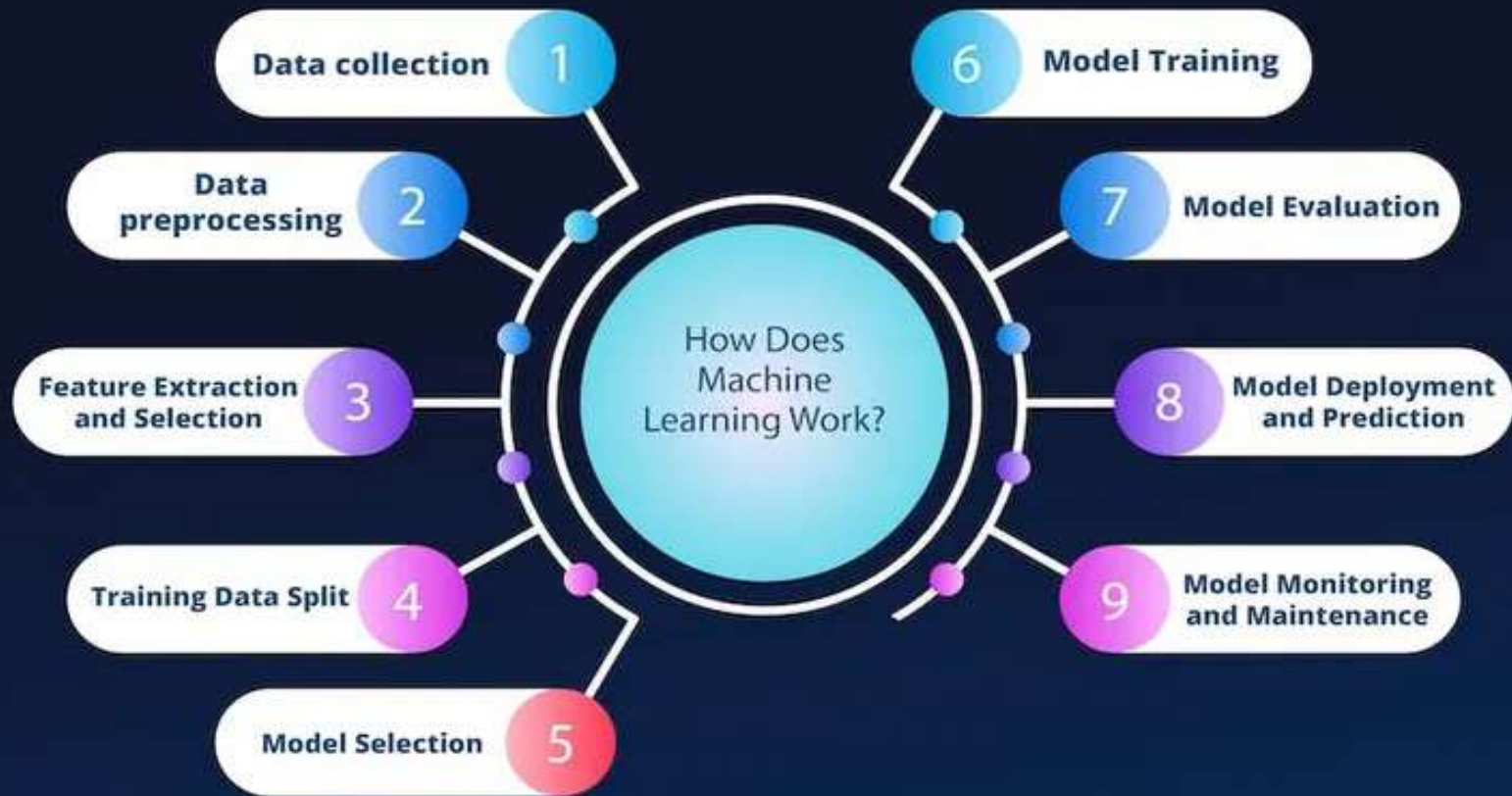
- **1. Solving Complex Business Problems:**
 - Image and speech recognition in healthcare.
 - Language translation and sentiment analysis.
- **2. Handling Large Volumes of Data:**
 - Fraud detection in financial transactions.
 - Personalized feed recommendations on Facebook and Instagram from billions of interactions.
- **3. Automate Repetitive Tasks:**
 - Gmail filtering spam emails automatically.
 - Chatbots handling order tracking and password resets.
 - Automating large-scale invoice analysis for key insights.

Why do we need Machine Learning?

- **4. Personalized User Experience**
 - Netflix suggesting movies and TV shows based on our viewing history.
 - E-commerce sites recommending products we're likely to buy.
- **5. Self Improvement in Performance**
 - Voice assistants like Siri and Alexa learning our preferences and accents.
 - Search engines refining results based on user interaction.

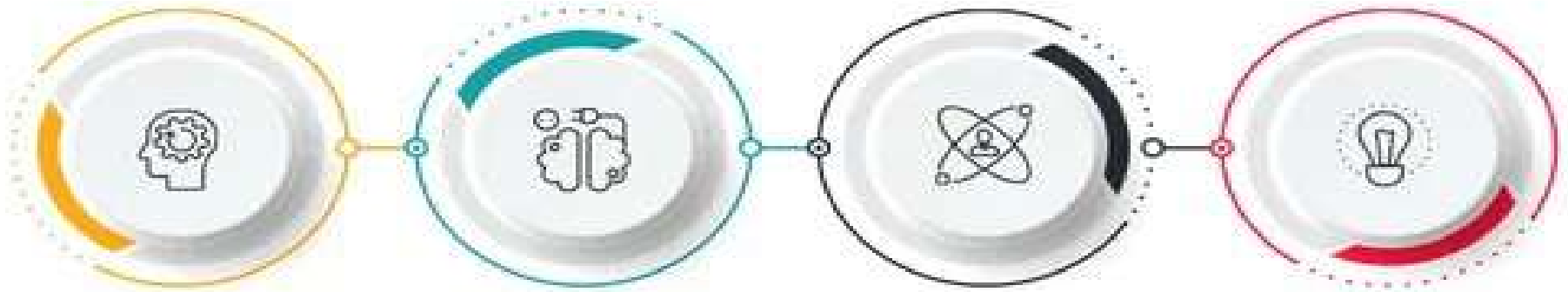
Key Concepts in Machine Learning

- **Training Data & Testing Data:** Training data is used to build the model, while testing data evaluates its performance.
- **Features & Labels:** Features are the input variables (e.g., age, height), and labels are the output or prediction (e.g., whether a person is healthy).
- **Model Training & Evaluation:** The model learns from training data, and its accuracy is tested on new data to ensure it generalizes well.
- **Optimization:** This process involves adjusting the model to minimize errors and improve accuracy.



Types of Machine Learning

TYPES OF MACHINE LEARNING



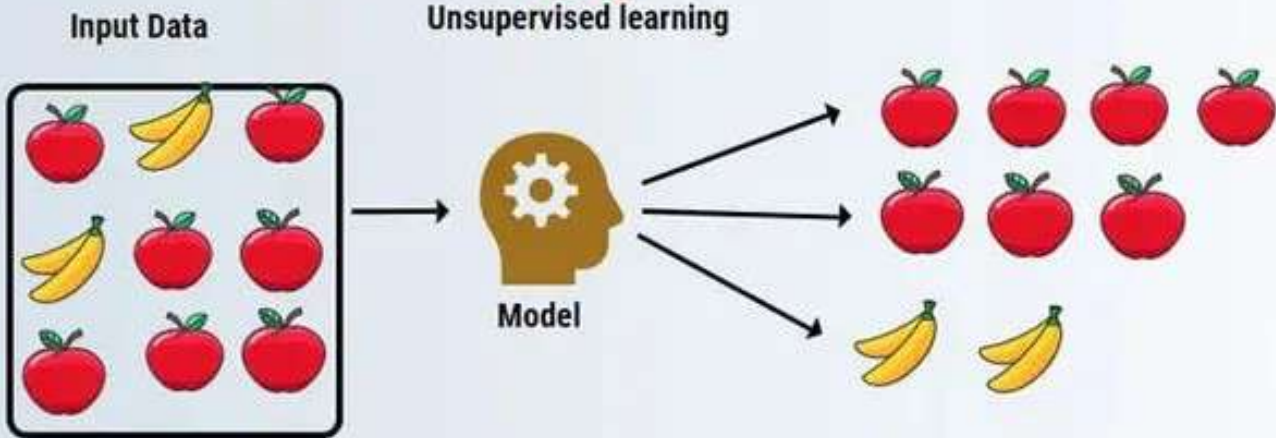
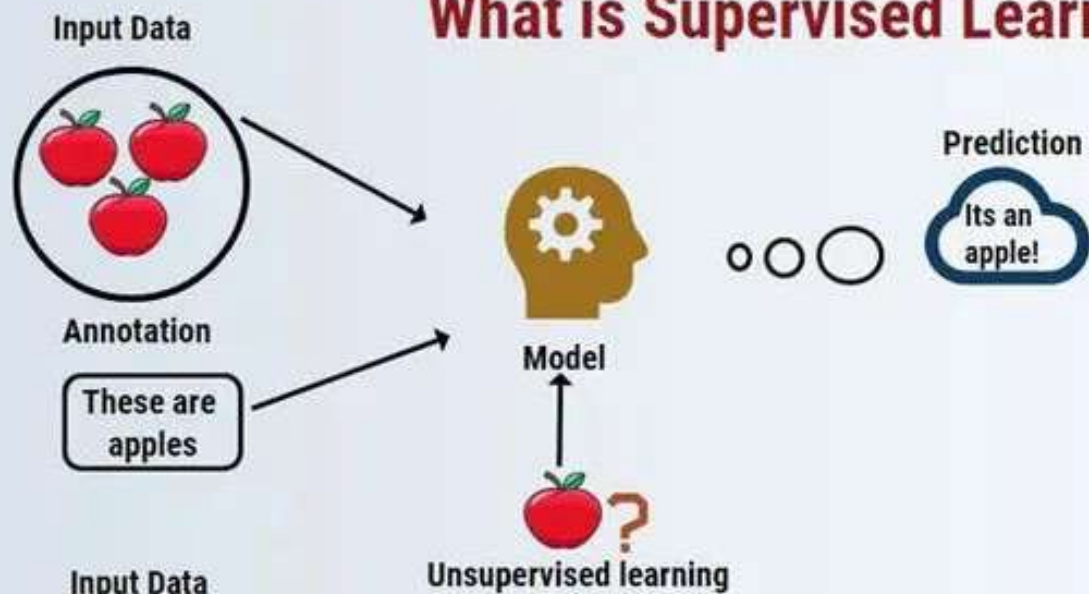
**Supervised
Machine Learning**

**Unsupervised
Machine Learning**

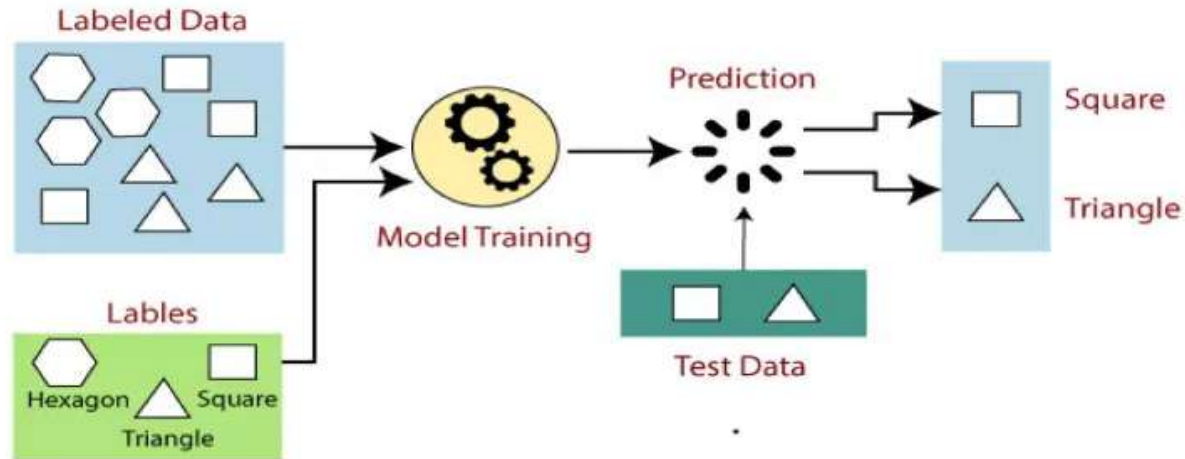
**Semi-Supervised
Learning**

**Reinforcement
Learning**

What is Supervised Learning?



How Supervised Learning Works?



Supervised machine learning

In supervised learning, the model is trained using data that already has correct labels.

This means the data comes with the right answers.

It is similar to learning with the help of a teacher or supervisor.

The algorithm learns from labeled data and can then predict answers for new, unseen data.

A major disadvantage is that labeling data takes a lot of time and effort, making it costly, especially for large datasets.

Example:

- **Simple Example:** Identifying Fruits
- Imagine you have a basket of fruits. Each fruit has a label: “Apple,” “Banana,” “Orange.” You show these labeled fruits to a computer:
- This is an apple → (round, red)
- This is a banana → (long, yellow)
- This is an orange → (round, orange)
- The computer learns from these examples. Later, if you give it a new fruit without a label, the computer will use what it learned and say:
- “This looks round and red... so it is an apple.”

In short:

- Supervised learning is like teaching a child using examples with correct answers.

Supervised Learning Use Cases

Some common use cases for supervised learning

Image classification: Identifying objects or features within an image, such as recognizing handwritten digits or determining whether an image contains a certain type of animal.

Text classification: Categorizing text into different classes, such as determining whether an email is spam or not.

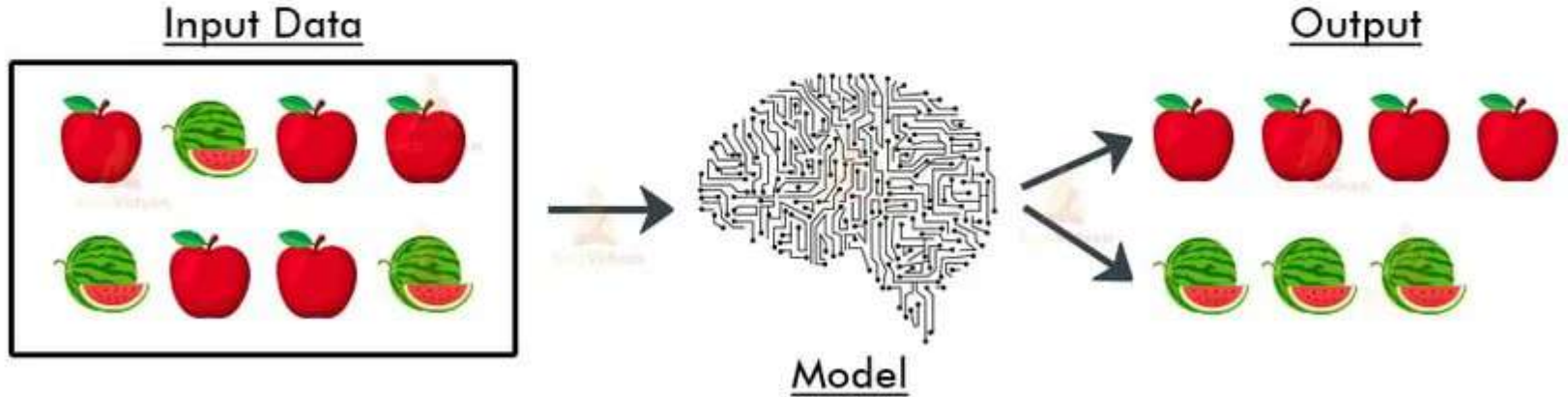
Natural Language Processing (NLP): Tasks such as language translation, text summarization, and sentiment analysis.

Predictive modeling: Using historical data to make predictions about future events, such as stock prices or weather patterns.

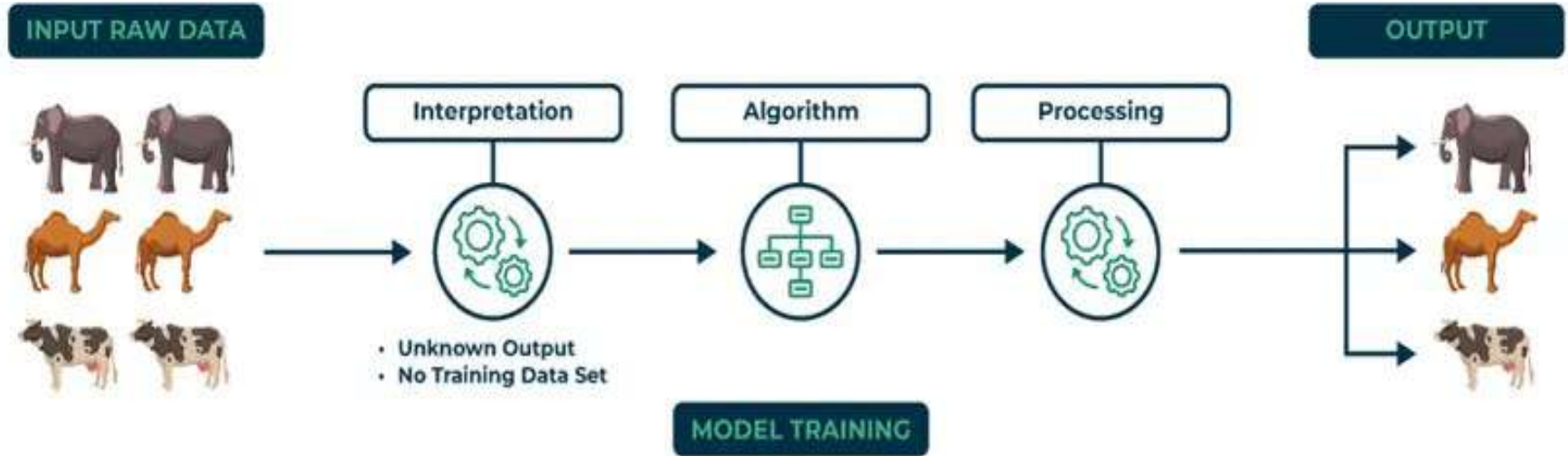
Medical diagnosis: Identifying diseases from medical images, ECG or other vital signs

Object detection in videos and images : Identify and locate objects in videos and images.

Unsupervised Learning in ML



Unsupervised Learning



Unsupervised machine learning

- In unsupervised learning, the model gets data without labels (no correct answers given).
- The model has to find patterns on its own.
- A common method is clustering, where similar items are grouped together.
- **Example:** If you give customer data, the model will group customers based on similar details like age, location, or spending habits.
- No teacher or supervisor is involved — the model discovers the structure by itself.

Example

Grouping Candies:

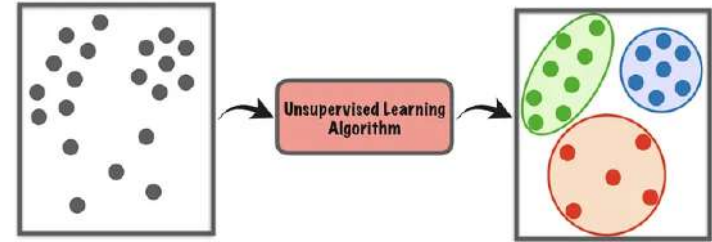
Imagine you have a big mix of candies, but none of them have labels. The computer looks at them and groups them based on similarities:

- All red candies in one group
- All round candies in another group
- All chocolate candies in another group

No one tells the computer the names of the candies — it just finds patterns and groups them on its own. What this shows:

- No labels
- No teacher
- Computer discovers patterns

Forms groups based on similar features



Unsupervised Learning Use Cases

Unsupervised learning is used for a variety of tasks, including:

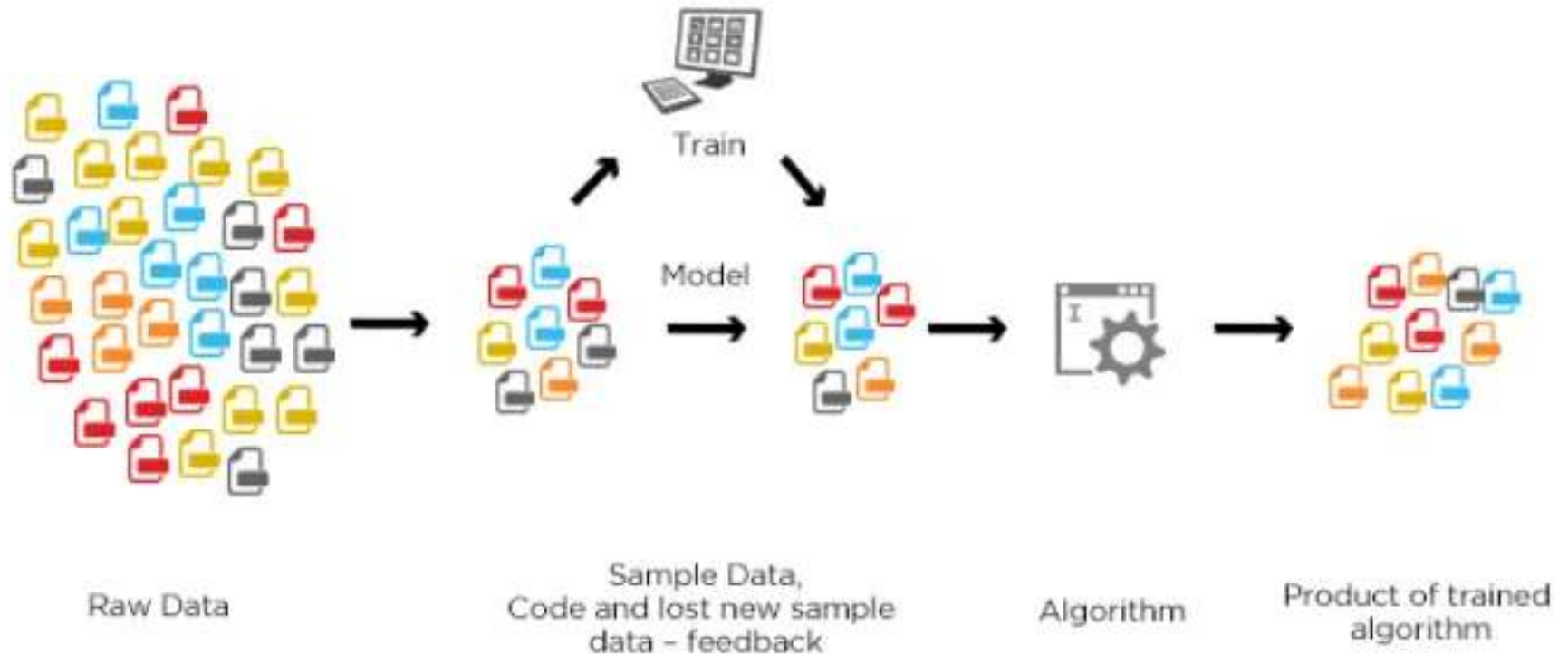
Clustering: Grouping similar data points together, such as grouping customers based on their purchasing patterns.

Dimensionality reduction: Reducing the number of features in a dataset, such as identifying the most important features in a set of images.

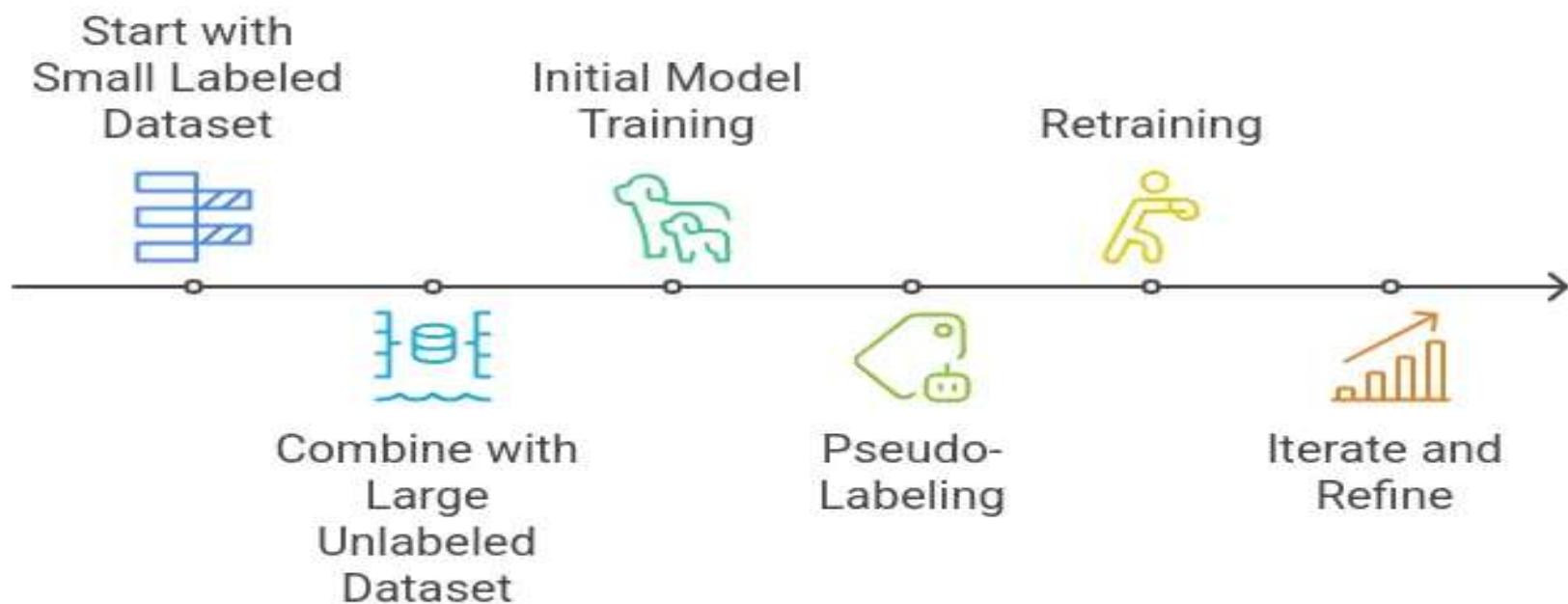
Anomaly detection: Identifying unusual or abnormal data points, such as detecting fraud in financial transactions.

Generative models: Creating new data that is similar to the input data, such as generating new images or text.

Semi-Supervised Learning



How does Semi-Supervised Learning Work



Semi-supervised Learning

- The model is trained using a small amount of labeled data (with correct answers)
- A large amount of unlabeled data (without answers).
- The few labeled examples help the model understand the data.
- The model then uses the unlabeled data to learn more patterns and improve accuracy.

Example

Imagine a teacher gives you:

- 5 homework questions with answers (labeled)
- 50 homework questions without answers (unlabeled)
- Students learn the method from the 5 answered examples and then try to solve the remaining 50 on your own using what you understood.
- This is semi-supervised learning.

Semisupervised Learning Usecase

1. Email Spam Detection

Only a few emails are marked as “spam” or “not spam,” but thousands of emails are unlabeled. The model learns from the few labeled emails and then uses the rest to improve spam detection.

2. Speech Recognition

Only some audio clips have written transcripts. Most audio data is unlabeled because labeling takes time. The model learns speech patterns from both labeled and unlabeled audio.

3. Identifying Fake News

Some articles are tagged as real or fake. Many more articles are unlabeled. Semi-supervised learning helps classify news using both types of data.

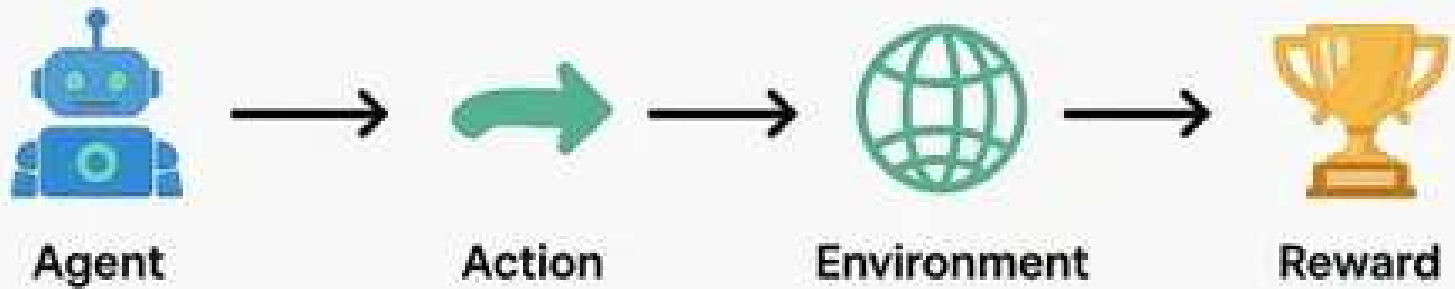
4. Medical Imaging (X-rays, MRI)

Few medical images are labeled by doctors. Large amounts of images are unlabeled. The model uses both to detect diseases more accurately.

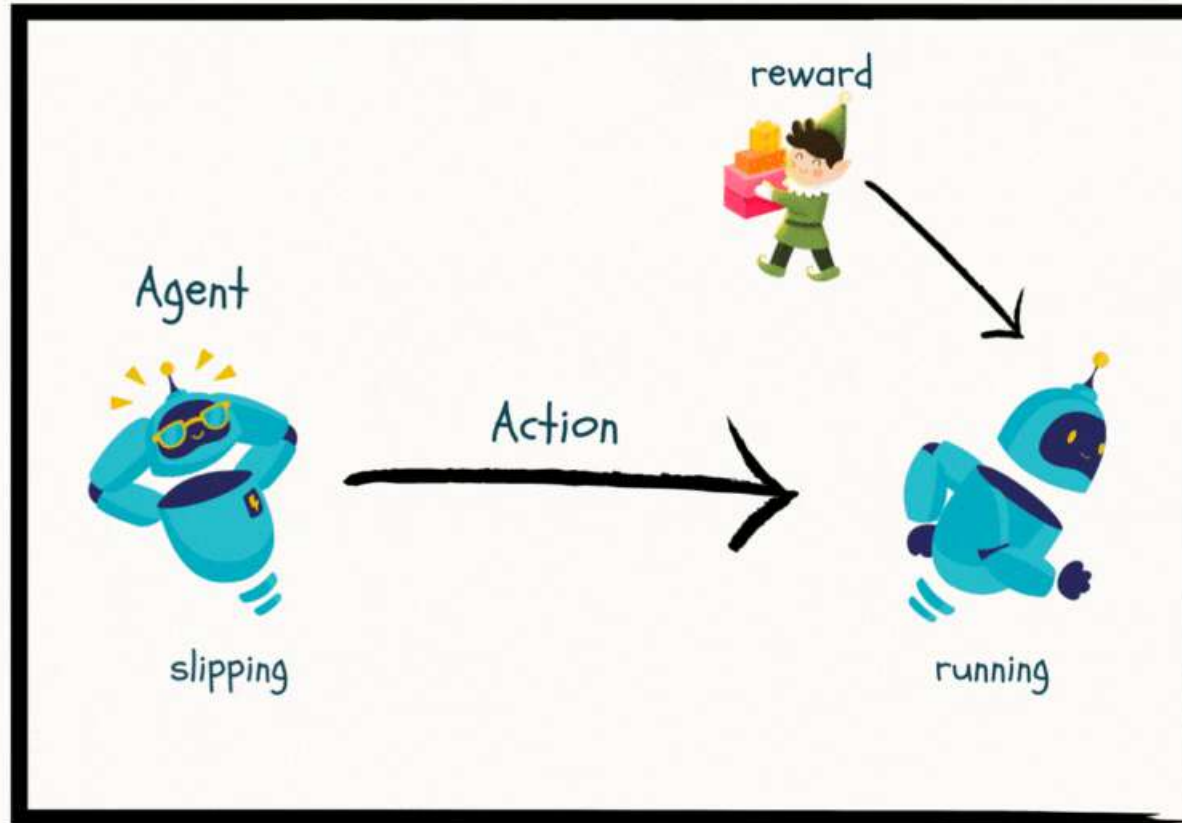
5. E-commerce Product Categorization

Only some products are labeled (e.g., Electronics, Fashion). Most new items have no category. The model uses semi-supervised learning to group and classify products.

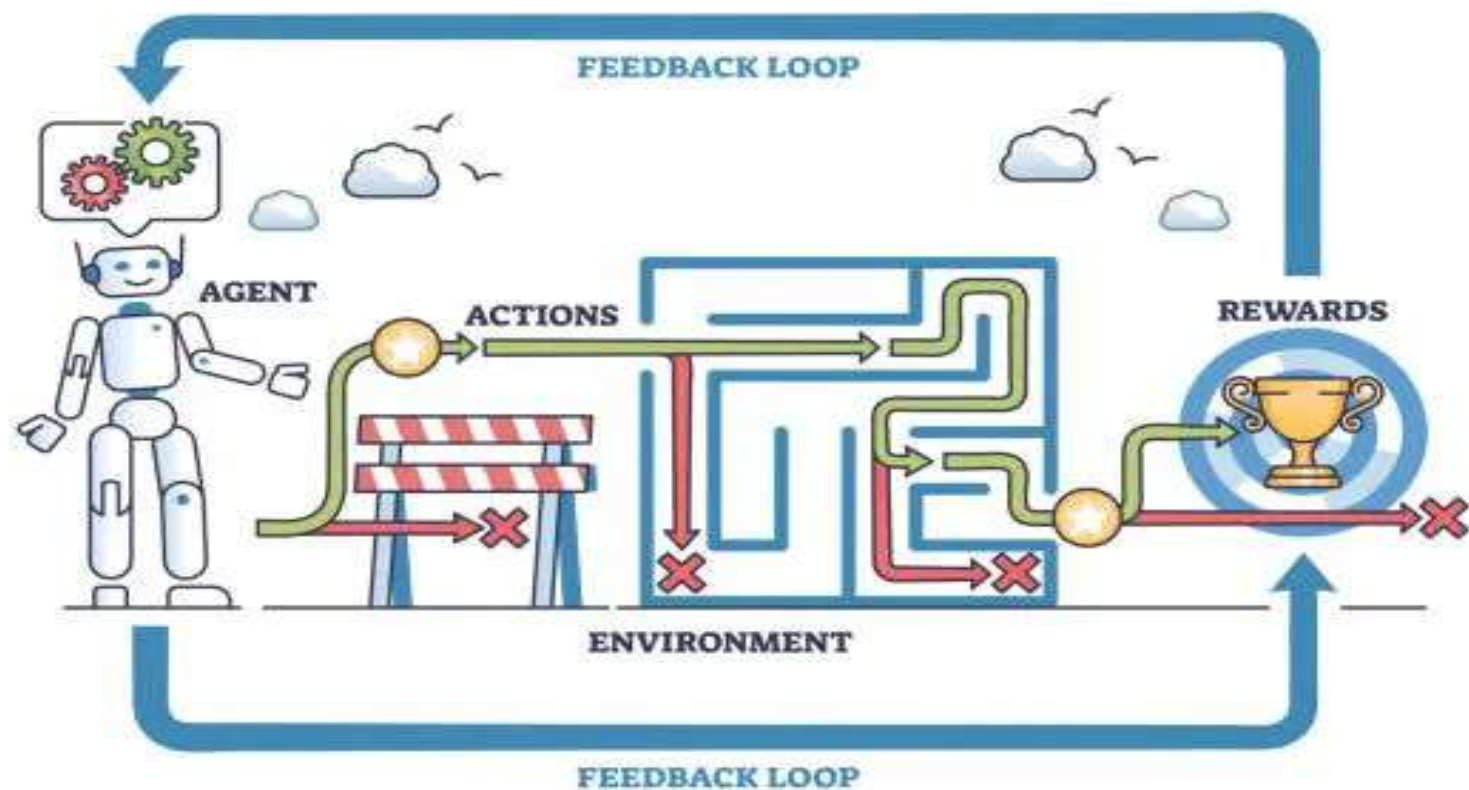
Reinforcement learning



Environment



REINFORCEMENT LEARNING



Reinforcement learning

- Reinforcement learning uses goal-oriented algorithms to achieve a complex objective.
- The agent learns to maximize rewards over a long period of time.
- No training dataset is used in reinforcement learning.
- This is different from supervised learning, where the model is trained using correct answers.
- In reinforcement learning, no correct answer is given to the agent.
- The agent learns from its own actions, experiences, rewards, and penalties.
- It decides what to do based on experience, not on labeled training data.
-
- **Positive reinforcement learning:** This refers to adding a reinforcing stimulus after a specific behavior of the agent, which makes it more likely that the behavior may occur again in the future, e.g., adding a reward after a behavior.
- **Negative reinforcement learning:** Negative reinforcement learning refers to strengthening a specific behavior that avoids a negative outcome.

Reinforcement Learning use cases

Reinforcement learning is used in a variety of tasks, including:

Robotics: Training robots to perform specific tasks, such as grasping objects or moving through a maze.

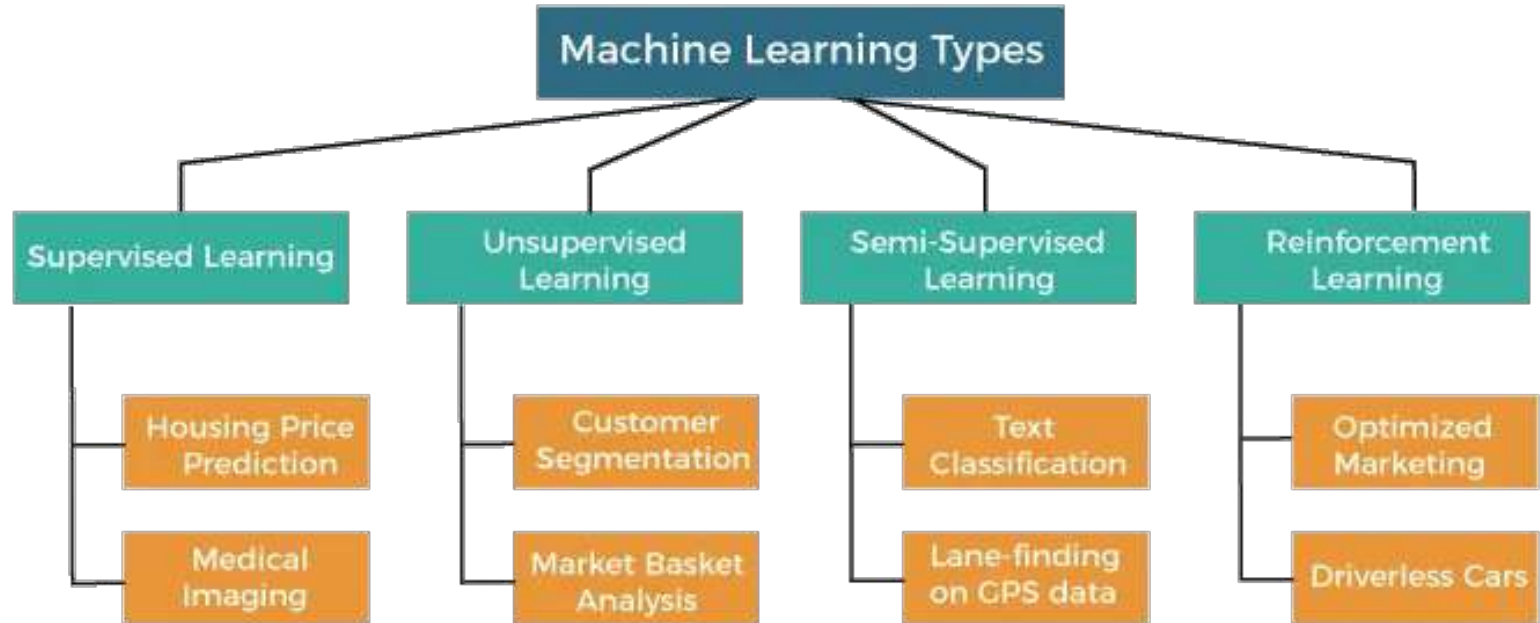
Game playing: Training agents to play games such as chess, Go, and poker at a superhuman level.

Autonomous vehicles: Training self-driving cars to navigate through traffic and make safe driving decisions.

Industrial control: Optimizing the control of industrial systems, such as power plants or manufacturing processes.

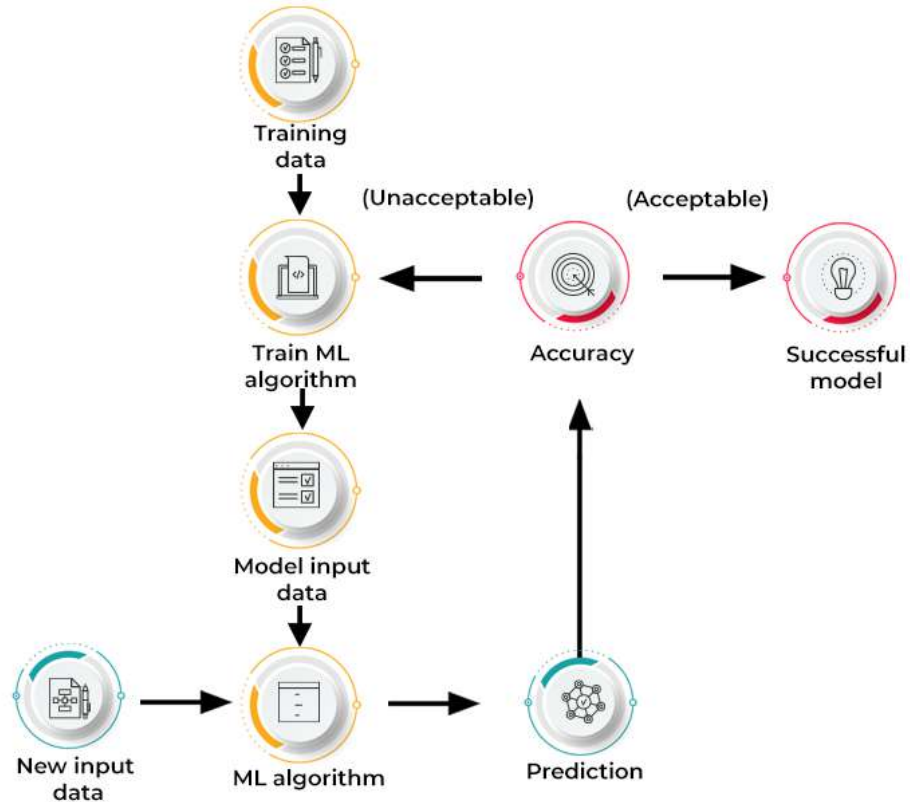
Healthcare: Developing personalized treatment plans for patients based on their medical history and response to treatment.

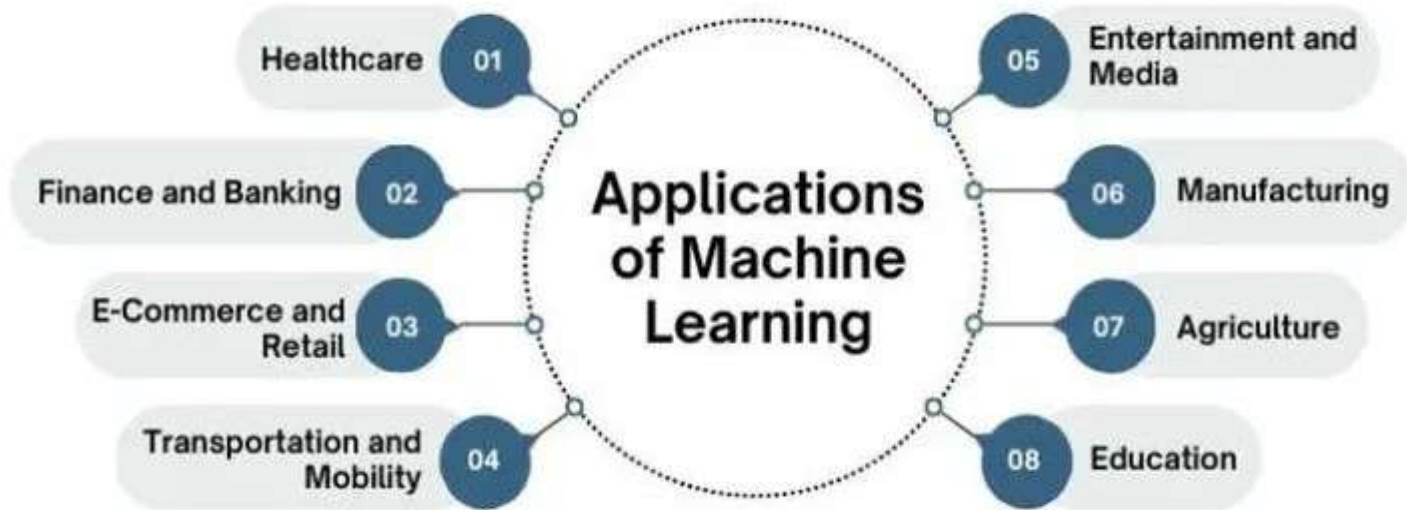
Finance: Developing trading strategies and portfolio management.



Working of ML

HOW DOES MACHINE LEARNING WORK?





Applications of Machine Learning

Virtual Assistants

Pattern Recognition

Face Recognition



Recommendation Engines

Fraud Detection

Speech Recognition

ML Applications

Online Fraud Detection - Machine learning is proving its potential to make cyberspace a secure place and tracking monetary frauds online is one of its examples. For example: Paypal is using ML for protection against money laundering.

Image Recognition – It is an approach for cataloging and detecting a feature or an object in the digital image. E.g. pattern recognition, face detection, or face recognition.

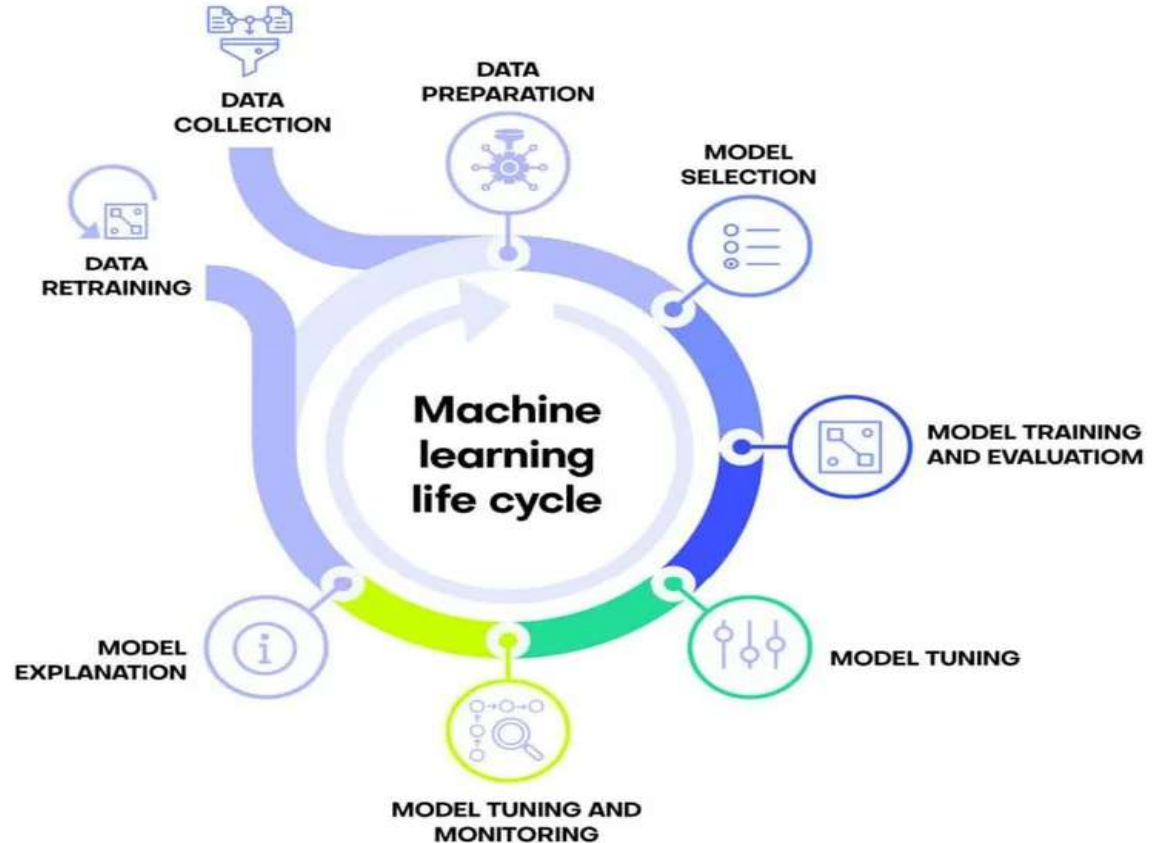
Virtual Personal Assistant - Siri, Alexa, Google Now are some of the popular examples of virtual personal assistants.

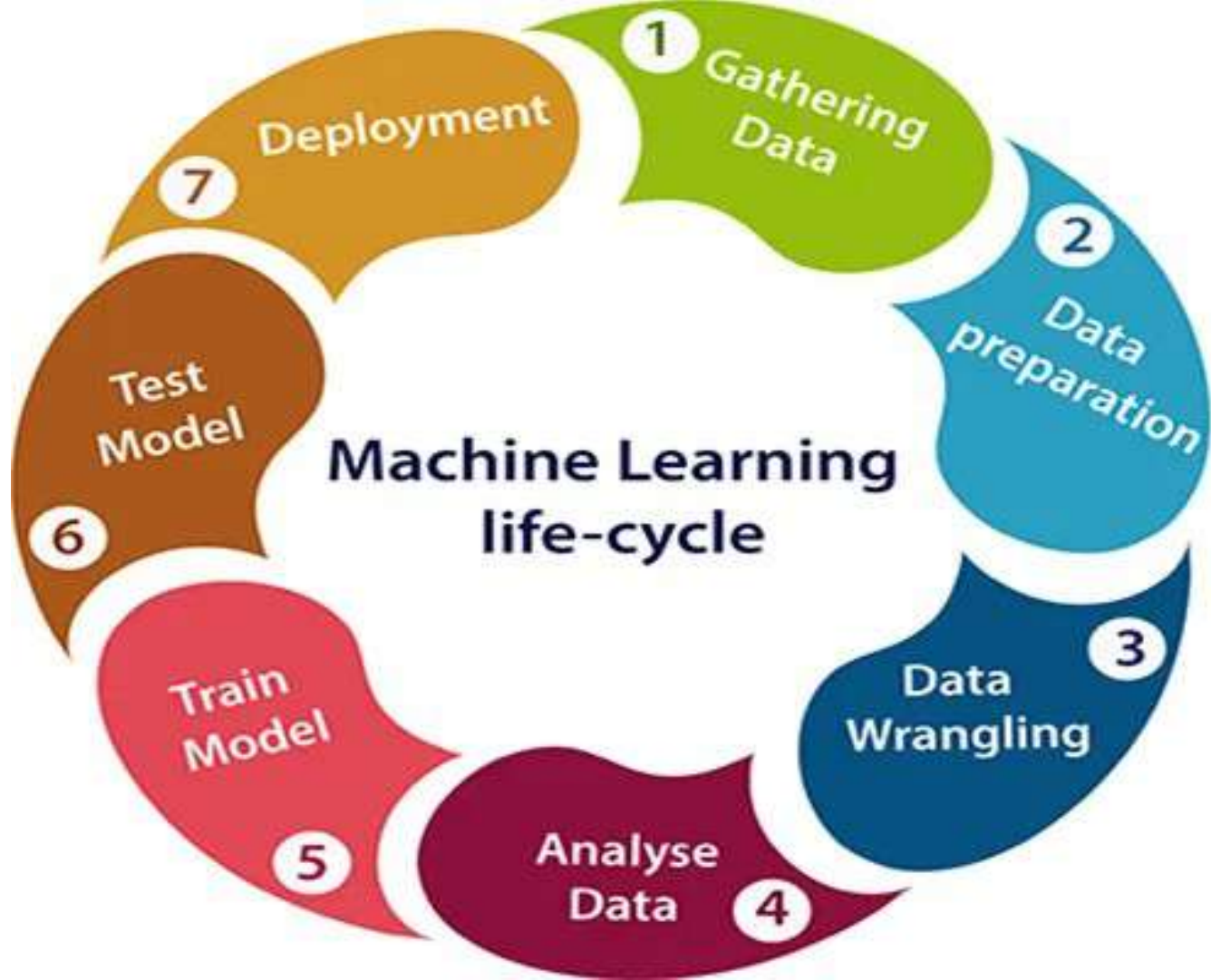
Email Spam and Malware Filtering - There are a number of spam filtering approaches that email clients use. To ascertain that these spam filters are continuously updated, they are powered by machine learning.

Product Recommendations - Product recommendation is one of the stark features of almost every e-commerce website today, which is an advanced application of machine learning techniques. Using machine learning and AI, websites track your behavior based on your previous purchase, your searching pattern, your cart history, and make product recommendations.

Machine Learning Life Cycle

Machine learning life cycle is a cyclic process to build an efficient machine learning project. The main purpose of the life cycle is to find a solution to the problem or project.





1. Problem Definition

Identify the problem you want to solve.

Understand the goal: classification, prediction, clustering, etc.

Define success metrics (accuracy, precision, error rate).

2. Data Collection

Identify the different data sources, as data can be collected from various sources such as **files**, **database**, **internet**, or **mobile devices**.

The **quantity and quality** of the collected data will determine the efficiency of the output. The more will be the data, the more accurate will be the prediction.

This step includes the below tasks:

- Gather relevant data from sources like databases, sensors, files, or APIs.
- Ensure sufficient quantity and quality of data.
- Data can be structured (tables) or unstructured (images, text).

Coherent set of data is also called as a **dataset**.

3. Data Preparation

- Prepare the data to use in machine learning training. This step can be further divided into two processes:
- **Data exploration:**
- It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data.
- A better understanding of data leads to an effective outcome.
- **Data pre-processing:**
 - preprocessing of data for its analysis.
 - Clean the data by removing errors, duplicates, and missing values.
 - Convert data into a suitable format (normalization, encoding).
 - Split data into training, validation, and test sets.

4. Data Wrangling

- It is the process of **cleaning and converting raw data into a useable format**.
- It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis
- Cleaning of data is required to address the quality issues. collected data may have various issues, including: **Missing Values**
- **Duplicate data Invalid data Noise**



#	Car Make	Car Model	Car Year	...	...	Sell Year	...
1	Toyota	Camry	2018	...	...	2018	...
2	Toyota	Corolla	2019	...	...	2019	...
3	Toyota	Camry	2018	...	...	2018	...
4	Toyota	Corolla	2019	...	...	2019	...
...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...
9999	Toyota	Camry	2018	...	...	2018	...
10000	Toyota	Camry	2018	...	...	2018	...

	col1	col2	col3	col4	col5			col1	col2	col3	col4	col5	
0	2	5.0	3.0	6	NaN	mean()		0	2.0	5.0	3.0	6.0	7.0
1	9	NaN	9.0	0	7.0			1	9.0	11.0	9.0	0.0	7.0
2	19	17.0	NaN	9	NaN			2	19.0	17.0	6.0	9.0	7.0

5. Data Analysis

- This step involves:
- **Selection of analytical techniques Building models**
- **Review the result**
- build a machine learning model to analyze the data using various analytical techniques and review the outcome.
- It starts with the determination of the type of the problems, where we select the machine learning techniques such as **Classification, Regression, Cluster analysis, Association**, etc. then build the model using prepared data, and evaluate the model.

6. Train Model

- Train model to improve its performance for better outcome of the problem.
- Training a model is required so that it can understand the various patterns, rules, and, features.
- Feed the training data to the selected algorithm.
- Model learns patterns and relationships in data.
- Tune hyperparameters to improve accuracy.

7. Test Model

- Machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it.
- Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.
- Test the model using validation/test data.
- Use metrics like accuracy, precision, recall, F1-score, RMSE.
- Check for issues like overfitting or underfitting.

8. Deployment

- The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.
- Deploy the trained model to real-world systems.
- Integrate it into applications, websites, or devices.
- Make predictions on live data.

Introduction to Data in Machine Learning

- Data is the foundation of Machine Learning (ML). Machine learning models learn patterns, make predictions, and take decisions based on the data provided to them. Without data, machine learning cannot exist.
- In ML, data acts like the experience from which a model learns—just as humans learn from past experiences. Data refers to the set of observations or measurements to train a machine learning models. The performance of such models is heavily influenced by both the quality and quantity of data available for training and testing.



What is Data?

Data is a collection of facts, observations, or measurements. In machine learning, data is usually represented in the form of:

- Numbers
- Text
- Images
- Audio
- Videos
- Sensor readings

This data helps the machine understand and learn relationships within a problem.

Properties of Data

- **Volume:** The scale of data generated every millisecond.
- **Variety:** Different data types like healthcare, images, videos, and audio.
- **Velocity:** The speed of data generation and streaming.
- **Value:** The meaningful insights data provides.
- **Veracity:** The accuracy and reliability of data.
- **Viability:** Data's adaptability for integration into systems.
- **Security:** Preventing tampering and unwanted access.
- **Accessibility:** Simple access for decision-making.
- **Integrity:** Accuracy and consistency throughout its lifecycle.
- **Usability:** Simplicity and interpretability for end-users.

Types of Data in Machine Learning

Types of Data

Structured Data

Data organized in rows and columns
e.g. spreadsheet and database



Unstructured Data

Not organized in a predefined manner (e.g., text, images, videos).



Semi-Structured Data

Has some organizational properties (e.g., JSON, XML).



1. Structured Data

- Structured data is highly organized and stored in predefined formats, such as tables, spreadsheets, and relational databases. It follows a fixed schema with clear relationships between data points.
 - Examples: Customer records in an SQL database, financial transactions in spreadsheets, or inventory management systems.
 - Applications: Used in predictive modeling, fraud detection, and business intelligence.

2. Unstructured Data

- Unstructured data lacks a predefined format and does not fit neatly into relational databases. It is typically complex, diverse, and requires advanced preprocessing before use in machine learning models.
 - Examples: Images, videos, audio files, emails, and social media posts.
 - Applications: Used in computer vision, sentiment analysis, and speech recognition. Deep learning models, such as CNNs (Convolutional Neural Networks) for image processing and NLP models for text processing, are commonly used for unstructured data.

3. Semi-Structured Data

- Semi-structured data has some level of organization but does not follow a strict schema like structured data. It contains tags or metadata that help define relationships.
 - Examples: JSON and XML files, NoSQL databases like MongoDB, and email metadata.
 - Applications: Used in web scraping, log analysis, and document classification.

Data Collection

Data collection is the first and one of the most important steps in the Machine Learning process. It involves gathering the raw information needed to train and evaluate a model. The quality and quantity of collected data directly affect the accuracy and performance of the machine learning system.

Why is Data Collection Important?

Helps the model learn correct patterns.

Determines how well the model will perform in real-world situations.

More and better data usually results in more accurate predictions.

Planning of Data Collection

Data can be collected from various places, such as:

- **Surveys or forms:** Structured questionnaires administered to a target group to gather specific information.
- **Interviews:** Conducting one-on-one or group conversations to gain in-depth insights.
- **Observation:** Systematically observing and recording behaviors or events.
- **Experiments:** Controlling variables to study cause-and-effect relationships.
- **Web scraping:** Extracting data from websites and online sources.
- **Sensor(IoT devices) data collection:** Gathering data from sensors or IoT devices.
- **Databases (student records, sales data)**
- **APIs (weather data, stock data)**
- **Logs and transactions**
- **Images, videos, audio recordings**

Methods of Data Collection

- **Manual Collection:** Humans collect data by observation, recording, or entering values.
- **Automated Collection:** Data is collected automatically using tools, scripts, sensors, or software.
- **Real-Time Data Collection:** Data comes continuously (e.g., traffic sensors, live financial data).
- **Batch Collection:** Data is collected and stored periodically (e.g., daily sales reports).

Important Considerations

- **Accuracy:** Data should be correct and reliable.
- **Completeness:** Avoid missing values.
- **Relevance:** Collect only data that is useful for the model.
- **Ethics & Privacy:** Follow legal rules while collecting user data.

Data Preprocessing

Data Pre-processing is the step in machine learning where raw data is cleaned, organized, and transformed into a format suitable for building a model. Real-world data is often incomplete, inconsistent, or contains errors. Pre-processing improves data quality and ensures better model performance.

Why is Data Pre-processing Important?

- Removes noise and errors
- Handles missing values
- Converts data into a usable and consistent format
- Improves model accuracy and reliability

Major Steps in Data Pre-processing

1. Data Cleaning:

Fixing or removing incorrect, duplicated, or inconsistent data.

Handling missing values (remove, replace, or fill)

2. Data Transformation:

Converting data into suitable formats.

Scaling and normalization (making values smaller and uniform).

Encoding categorical data (e.g., converting “Male/Female” to numbers)

3. Data Integration:

Combining data from multiple sources

Ensuring consistency across merged datasets

Major Steps in Data Pre-processing

4. Data Reduction

Reducing the amount of data while keeping important information

Techniques: feature selection, dimensionality reduction

5. Data Splitting

Dividing data into:

Training set

Validation set

Test set

Functions used for data pre-processing

Data pre-processing involves many built-in functions to clean, transform, and prepare data for model building. These functions help handle missing values, encode categories, scale numerical data, and detect errors.

How Data is Split in Machine Learning?

- **Training data** is the largest portion of the dataset, used to train the model by adjusting its parameters based on patterns in the data.
- **Validation data** is a separate subset used to fine-tune model parameters and prevent overfitting. It helps in selecting the best model configurations.
- **Test data** is used after training to evaluate the model's performance on unseen data, ensuring it generalizes well to real-world applications.